Pokhara University
Faculty of Science and Technology

Course Code: CMP 382
Course Title: **Cloud Computing**
Nature of the course: Theory + Practical
Year: 3, Semester VI
Level: Bachelor
Program: **BCSIT**

Full marks: 100
Pass marks: 45
Credit Hours: 3
Total Periods: 48 hrs.

1. Course Description

This course aims to provide basic concepts of cloud computing such as cloud architecture, deployment models and cloud service management. It is designed to provide students with a comprehensive understanding of how cloud platforms operate and how computing resources are delivered over the internet.

Students will gain practical skills required to design, deploy and manage applications in cloud environments. It also covers key areas such as virtualization, storage systems, networking, security, and cost optimization in cloud platforms.

At the end of this course, students will be able to understand cloud computing fundamentals and apply cloud-based solutions effectively in real-world scenarios.

2. Course Objectives

Through this course, students shall be able:

- To understand the fundamental concepts, evolution, and real-world applications of cloud computing
- To analyze and apply different cloud service and deployment models based on requirements
- To understand cloud architecture, including scalability, distributed systems, and modern design approaches
- To work with core technologies such as virtualization and containerization for efficient resource management
- To design, deploy, and manage basic cloud-based applications and services
- To understand distributed computing concepts and basic big data processing techniques
- To identify and address cloud security, privacy, and governance issues
- To gain practical experience using cloud tools, platforms, and development environments

3. Method of Instructions

General instructional techniques: Lecture, discussion, demonstrations and interactive question answer sessions
Specific instructional techniques: Practical implementation using tools such as virtualization software, Docker and cloud platforms.

4. Course Contents

Specific Objectives	Contents
Unit 1: Introduction to Cloud Computing [5 Hrs.]	
● Explain the concept, evolution and key characteristics of cloud computing ● Describe the benefits, challenges and real-world applications of cloud computing ● Identify cloud storage concepts and infrastructure requirements ● Explain cloud adoption strategies and dynamic infrastructure ● Analyze real-world case studies of cloud adoption	1.1 Cloud computing overview and evolution 1.2 Characteristics of Cloud Computing 1.3 Benefits and challenges 1.4 Applications of cloud computing 1.5 Introduction to cloud storage 1.6 Cloud service requirements 1.7 Dynamic infrastructure 1.8 Cloud adoption strategies and case studies
Unit 2: Cloud Service and Deployment Models [7 Hrs.]	
● Differentiate and analyze cloud service models ● Compare cloud deployment models and evaluate their use cases ● Explain the shared responsibility model in cloud environments ● Evaluate appropriate service and deployment models based on given scenarios ● Explain multi-cloud concepts and vendor lock-in issues ● Describe serverless computing and event-driven architecture	2.1 Cloud service models: IaaS, PaaS, SaaS 2.2 Deployment models: public, private, hybrid, community 2.3 Comparison of service models 2.4 Shared responsibility model 2.5 Selection criteria 2.6 Multi Cloud concepts 2.7 Vendor lock-in issues 2.8 Serverless computing and event-driven architecture
Unit 3: Cloud Computing Architecture [6 Hrs.]	
● Explain cloud architecture and its core components ● Compare monolithic and microservices architectures	3.1 Layered cloud architecture 3.2 Service-Oriented Architecture (SOA) 3.3 Microservices vs monolithic architecture 3.4 Scalability and elasticity, fault tolerance

• Apply concepts of scalability, elasticity and fault tolerance • Explain high availability and load balancing mechanisms • Analyze security, trust and privacy in cloud architecture • Relate architectural concepts to real-world cloud systems	and high availability 3.5 Load distribution concepts 3.6 Security, trust and privacy considerations 3.7 Case studies of cloud architectures
Unit 4: Cloud Applications and Service Management [8 Hrs.]	
• Explain design principles of cloud-based applications • Apply techniques for cloud application deployment and migration • Analyze cloud cost models and pricing strategies • Explain service level agreements, performance, and availability metrics • Analyze portability and interoperability in cloud systems • Apply basic DevOps practices in cloud environments • Describe monitoring and logging mechanisms in cloud systems	4.1 Design principles of cloud applications 4.2 Web application architecture in cloud 4.3 Application migration and software licensing in cloud 4.4 Cost models and pricing strategies 4.5 Service Level Agreements (SLA) 4.6 Performance and availability metrics 4.7 Portability and interoperability 4.8 Introduction to DevOps and CI/CD concepts 4.9 Infrastructure as Code (IaC) basics 4.10 Monitoring and logging in cloud systems
Unit 5: Virtualization and Containerization Technology [8 Hrs.]	
• Explain virtualization concepts and different types of virtualizations • Describe hypervisors and the lifecycle of virtual machines • Compare virtual machines and containers • Apply containerization concepts in practical scenarios • Explain serverless computing and event driven functions. • Explain load balancing and its concepts	5.1 Introduction to virtualization 5.2 Types: server, storage, network virtualization 5.3 Virtualization architecture and implementation 5.4 Hypervisors and types 5.5 Virtual machine lifecycle 5.6 Introduction to containerization 5.7 VM vs containers 5.8 Introduction to Serverless Computing 5.9 Event-driven functions 5.10 Load balancing concepts 5.11 Serverless vs. Containerized workloads
Unit 6: Distributed Computing and Big Data Processing [7 Hrs.]	

● Explain concepts of parallel and distributed computing ● Describe distributed file systems and data consistency mechanisms ● Apply the MapReduce model for data processing tasks ● Analyze big data processing techniques and frameworks ● Identify modern data processing models Relate distributed computing concepts to real-world applications	6.1 Parallel and distributed computing fundamentals 6.2 Distributed file systems 6.3 Data replication and consistency 6.4 MapReduce model and phases 6.5 Applications of MapReduce 6.6 Parallel efficiency 6.7 Batch processing techniques 6.8 Hadoop and MapReduce 6.9 Modern data processing models 6.10 Case studies in distributed systems
Unit 7: Cloud Security and Governance [7 Hrs.]	
● Identify cloud security risks and challenges ● Explain identity and access management mechanisms ● Apply data security and encryption techniques ● Analyze multi-tenancy and virtual machine security issues ● Explain legal, regulatory and governance frameworks ● Evaluate cloud security practices in real-world scenarios	7.1 Cloud security issues and challenges 7.2 Identity and access management (IAM) 7.3 Data security and encryption 7.4 Application and VM security 7.5 Multi Tenancy issues 7.6 Security monitoring and auditing 7.7 Legal and regulatory aspects 7.8 Data privacy and compliance 7.9 Governance models 7.10 Security case studies

5. **Practical works:**
   1. Install VirtualBox/VMware and create a virtual machine then observe resource allocation and VM behavior.
   2. Perform file upload, download, sharing and access control in a cloud storage system or simulated environment.
   3. Deploy a simple web application in a cloud environment.
   4. Monitor CPU, and other metrics of virtual machines.
   5. Calculate simple cloud costs using sample pricing models.
   6. Apply basic data encryption to understand cloud security practices.
   7. Implement basic Infrastructure as Code (IaC) scripts to deploy a cloud application.
   8. Explore a lightweight serverless function (e.g., AWS Lambda).
   9. Use Docker to run a simple containerized application and observe container behavior.

10. Implement a simple MapReduce program (such as word count) using a programming language to understand distributed processing.

6. Evaluation system and Student's Responsibility

In addition to the formal exam, the internal evaluation of a student may consist of assignments, lab reports, projects, class participation etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%	3		
Assignments	20%	6		
Presentations/Quizzes	10%	3		
Internal Assessment	60%	18		
Practical		**20**		
Attendance & Class Participation	10%	2		
Lab Report/Project Report	20%	4		
Practical Exam/Project Work	40%	8		
VIVA	30%	6		
Total Internal		**50**		
Full Marks: 50+50 = 100				

Student's Requirements:

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with a minimum of 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period.

Prescribed Books and References

1. Saurabh, K. (2011). *Cloud computing: Insights into new-era infrastructure*. Wiley India.
2. Buyya, R., Vecchiola, C., & Thamarai Selvi, S. (2013). *Mastering cloud computing: Foundations and applications programming*. Morgan Kaufmann.
3. Hurwitz, J., Bloor, R., Kaufman, M., & Halper, F. (2010). *Cloud computing for dummies*. Wiley Publishing.
4. Linthicum, D. S. (2009). *Cloud computing and SOA convergence in your enterprise: A step-by-step guide*. Addison-Wesley.
5. Sosinsky, B. (2011). *Cloud computing bible*. Wiley Publishing.
6. Erl, T., & Barcelo Monroy, E. (2019). *Cloud computing: Concepts, technology & architecture* (2nd ed.). Prentice Hall.